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AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges

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ABSTRACT

Information fusion, in the context of the Generative AI era, must distinguish AI Agents from Agentic AI. This review critically distinguishes between AI Agents and Agentic AI, offering a structured, conceptual taxonomy, application mapping, and analysis of opportunities and challenges to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven and enabled by LLMs and LIMs for task-specific automation. Generative AI is positioned as a precursor providing the foundation, with AI agents advancing through tool integration, prompt engineering, and reasoning enhancements. We then characterize Agentic AI systems, which, in contrast to AI Agents, represent a paradigm shift marked by multi-agent collaboration, dynamic task decomposition, persistent memory, and coordinated autonomy. Through a chronological evaluation of architectural evolution, operational mechanisms, interaction styles, and autonomy levels, we present a comparative analysis across both AI agents and agentic AI paradigms. Application domains enabled by AI Agents such as customer support, scheduling, and data summarization are then contrasted with Agentic AI deployments in research automation, robotic coordination, and medical decision support. We further examine unique challenges in each paradigm including hallucination, brittleness, emergent behavior, and coordination failure, and propose targeted solutions such as ReAct loops, retrieval-augmented generation (RAG), automation coordination layers, and causal modeling. This work aims to provide a roadmap for developing robust, scalable, and explainable AI-driven systems.

1. Introduction

Prior to the widespread adoption of AI Agents and Agentic AI (Fig. 1) around 2022 (Before ChatGPT was introduced), the development of autonomous and intelligent agents was deeply rooted in foundational paradigms of artificial intelligence, particularly multi-agent systems (MAS) and expert systems, which emphasized social action and distributed intelligence [1,2]. Notably, Castelfranchi [3] laid the critical groundwork by introducing ontological categories for social action, structure, and mind, arguing that sociality emerges from individual agents' actions and cognitive processes in a shared environment, with concepts like goal delegation and adoption forming the basis for cooperation and organizational behavior. Similarly, Ferber [4] provided a comprehensive framework for MAS, defining agents as entities with autonomy, perception, and communication capabilities, and highlighting their applications in distributed problem-solving, collaborative robotics, and synthetic world simulations.

These early studies established that individual social actions and cognitive architectures are fundamental to modeling collective phenomena, setting the stage for modern AI Agents. This paper builds on these foundational concepts to explore how social action modeling, as proposed in [3,4], informs the design of AI Agents capable of complex, socially intelligent interactions in dynamic environments.

Classical Agent-like systems were designed to perform specific tasks with predefined rules, which offered limited autonomy, and minimal adaptability to dynamic environments. These systems were primarily reactive or deliberative, relying on symbolic reasoning, rule-based logic, or scripted behaviors rather than the learning-driven, context-aware capabilities of modern AI Agents [5,6]. For instance, expert systems used knowledge bases and inference engines to emulate human decision-making in domains like medical diagnosis (e.g., MYCIN [7]). Other notable examples include DENDRAL [8], an expert system for molecular structure prediction; XCON [9], used for computer system

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configuration; and CLIPS [10], a rule-based production system framework. Systems like SOAR [11] and the subsumption architecture [12] extended symbolic and reactive logic into cognitive modeling and robotics.

In addition to task-specific reasoning, these agents supported limited forms of social interaction. Early conversational systems like ELIZA [13] and PARRY [14] simulated basic dialogue through pattern matching and script-based responses but lacked genuine understanding or contextual adaptation. Similarly, reactive agents in robotics executed sense-act cycles based on fixed control rules, as seen in early autonomous platforms like the Stanford Cart [15].

Multi-agent systems facilitated coordination among distributed entities, exemplified by auction-based resource allocation in supply chain management [16–18]. Scripted AI in video games, like NPC behaviors in early RPGs, used predefined decision trees [19]. Furthermore, BDI (Belief-Desire-Intention) architectures enabled goal-directed behavior in software agents, such as those in air traffic control simulations [20, 21].

However, across these diverse systems, early AI agents shared common limitations: they lacked self-learning, generative reasoning, and adaptability to unstructured or evolving environments. These shortcomings distinguish them from Agentic AI a recent paradigm that builds on deep learning, reinforcement learning, and foundation models to enable agents with contextual awareness, continuous learning, and emergent autonomy [22].

Recent public, academic and industry interest in AI Agents and Agentic AI reflects this broader transition in system capabilities. As illustrated in Fig. 1, Google Trends data demonstrates a significant rise in global search for both terms following the emergence of large-scale generative models in late 2022. This shift is closely tied to the evolution of agent design from the pre-2022 era, where AI Agents operated in constrained, rule-based environments, to the post-LLM period marked by learning-driven, flexible/adaptive architectures [23–25]. These newer systems enable agents to refine their performance over time and interact autonomously with unstructured, dynamic inputs [26–28]. For instance, while pre-modern expert systems required manual updates to static knowledge bases, modern agents leverage emergent neural architectures to generalize across tasks [25]. The surge in trend activity reflects growing awareness of this technological leap, as researchers and practitioners seek tools that go beyond automation toward autonomy and general-purpose reasoning. Moreover, applications are no longer confined to narrow domains like simulations or logistics, but now extend to broad practical settings demanding real-time reasoning and adaptive control. This momentum, as visualized in Fig. 1, highlights the significance of recent architectural advances in scaling autonomous agents for real-world deployment.

The release of ChatGPT in November 2022 marked a pivotal inflection point in the development and public perception of artificial intelligence, catalyzing a global surge in adoption, investment, and research activity [29]. In the wake of this breakthrough, the AI landscape underwent a rapid transformation, shifting from the use of standalone LLMs toward more autonomous, task-oriented frameworks [30]. This evolution progressed through two major post-generative phases: AI Agents and Agentic AI. Initially, the widespread success of ChatGPT popularized Generative Agents, which are LLM-based systems designed to produce novel outputs such as text, images, and code from user prompts [31,32]. These agents were quickly adopted across applications ranging from conversational assistants (e.g., GitHub Copilot [33]) and content-generation platforms (e.g., Jasper [34]) to creative tools (e.g., Midjourney [35]), revolutionizing domains like digital design, marketing, and software prototyping throughout 2023 and beyond.

Although the term AI Agent was first introduced in 1998 [3], it has since evolved significantly with the rise of generative AI. Building upon this generative foundation, a new class of systems commonly referred to as AI Agents has emerged. These agents enhanced LLMs with capabilities for external tool use (e.g., API-based tools), function

calling, and sequential reasoning, enabling them to retrieve real-time information and execute multi-step workflows autonomously [36,37]. Example frameworks such as AutoGPT [38] and BabyAGI (<https://github.com/yoheinakajima/babyagi>) highlight this transition, showcasing how LLMs could be embedded within feedback loops to dynamically plan, act, and adapt in goal-driven environments [39,40]. By late 2023, the field had advanced further into the realm of Agentic AI complex, multi-agent systems in which specialized agents collaboratively decompose goals, communicate, and coordinate toward shared objectives. In line with this evolution, Google introduced the Agent-to-Agent (A2A) protocol in 2025 [41], a proposed standard designed to enable seamless interoperability among agents across different frameworks and vendors. The protocol is built around five core principles: embracing agentic capabilities, building on existing standards, securing interactions by default, supporting long-running tasks, and ensuring modality agnosticism. These guidelines aim to lay the groundwork for a responsive, scalable agentic infrastructure.

Architectures such as CrewAI demonstrate how these agentic frameworks can accomplish decision-making across distributed roles, facilitating intelligent behavior in high-stake applications including robotics, logistics management, and adaptive decision-support [42].

As the field progresses from Generative Agents toward increasingly autonomous systems of Agentic AI, it becomes critically important to delineate the technological and conceptual boundaries between AI Agents and Agentic AI. While both paradigms build upon LLMs and extend the capabilities of generative systems, they embody fundamentally different architectures, interaction models, and levels of autonomy. AI Agents are typically designed as single-entity systems that perform goal-directed tasks by utilizing external tools, applying sequential reasoning, and integrating real-time information to complete well-defined functions [25,43]. In contrast, Agentic AI systems are composed of multiple, specialized agents that coordinate, communicate, and dynamically allocate sub-tasks within a broader workflow to achieve a common goal(s) [22,44]. This architectural distinction highlights clear and significant differences in scalability, adaptability, and application scope.

Understanding and formalizing the taxonomy between these two paradigms (**AI Agents and Agentic AI**) is scientifically significant for several reasons. First, it enables more precise system design by aligning computational frameworks with problem complexity ensuring that AI Agents are deployed for modular, tool-assisted tasks, while Agentic AI is employed for orchestrated multi-agent operations. Moreover, it allows for appropriate benchmarking and evaluation: performance metrics, safety protocols, and resource requirements differ substantially between agents designed for carrying out individual tasks and system of distributed agents designed for accomplishing complex, coordinated tasks. Additionally, clear taxonomy reduces development inefficiencies by preventing the misapplication of design principles such as assuming inter-agent collaboration in a system designed for single-agent execution. Without this clarity, developers and practitioners risk both under-engineering complex scenarios that require agentic coordination and over-engineering simple applications that could be solved with a single AI Agent.

This article aims to clarify the differences between AI Agents and Agentic AI, providing researchers with a foundational understanding of these technologies. The objective of this study is to formalize the distinctions, establish a shared vocabulary, and provide a structured taxonomy between AI Agents and Agentic AI that informs the next generation of intelligent agent design across academic and industrial domains, as illustrated in Fig. 2.

This review also provides a comprehensive conceptual and architectural analysis of the progression from traditional AI Agents to emergent Agentic AI systems. Rather than organizing the study around formal research questions common in review articles, we adopt a sequential, layered structure that clearly lays out the historical and technical evolution of these paradigms. Beginning with a detailed description

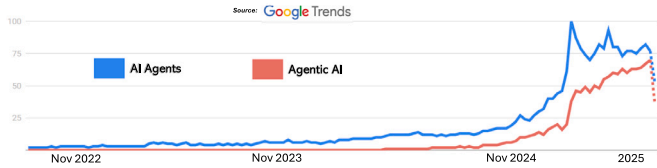


Fig. 1. Global Google search trends showing rising interest in “AI Agents” and “Agentic AI” since November 2022 when the ChatGPT was first introduced.

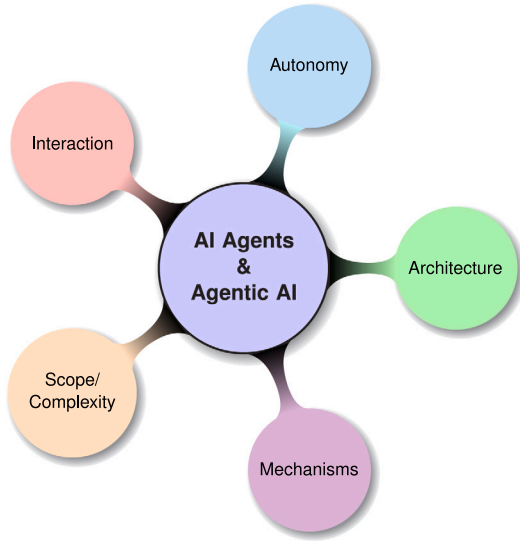


Fig. 2. Mind map of this research for exploring concepts, applications and challenges of AI Agents and Agentic AI. Each color-coded branch represents a key dimension of comparison: Architecture, Mechanisms, Scope/Complexity, Interaction, and Autonomy.

of our search strategy and selection criteria, we first establish the foundational understanding of AI Agents by analyzing their defining attributes, such as autonomy, reactivity, and tool-based execution. We then explore the critical role of foundational models, specifically LLMs and Large Image Models (LIMs), which serve as the core reasoning and perceptual engines that drive agentic behavior. Subsequent sections examine how generative AI systems have served as precursors to more dynamic, interactive agents, setting the stage for the emergence of Agentic AI. Through this perspective, we examine and present the conceptual leap from isolated, single-agent systems to orchestrated multi-agent architectures, highlighting their structural distinctions, coordination strategies, and collaborative mechanisms. We further map the architectural evolution by analyzing the core system components of both AI Agents and Agentic AI, offering comparative description of their planning, memory, orchestration, and execution layers. Building on this foundation, we review application domains spanning customer support, healthcare, research automation tasks, and robotics, while categorizing real-world deployments by system capabilities and coordination complexity. We then assess key challenges faced by both paradigms including hallucination, limited reasoning depth, causality deficits, scalability issues, and governance risks. To address these limitations, we outline opportunities for emerging solutions such as retrieval-augmented generation, tool-based reasoning, memory architectures, and simulation-based planning. The review concludes with a forward-looking roadmap that envisions the convergence of modular AI Agents and orchestrated Agentic AI in mission-critical domains such as autonomous vehicles, finance, and healthcare, and beyond. We aim to provide researchers with a structured taxonomy and actionable insights to guide the design, deployment, and evaluation of next-generation agentic AI systems.

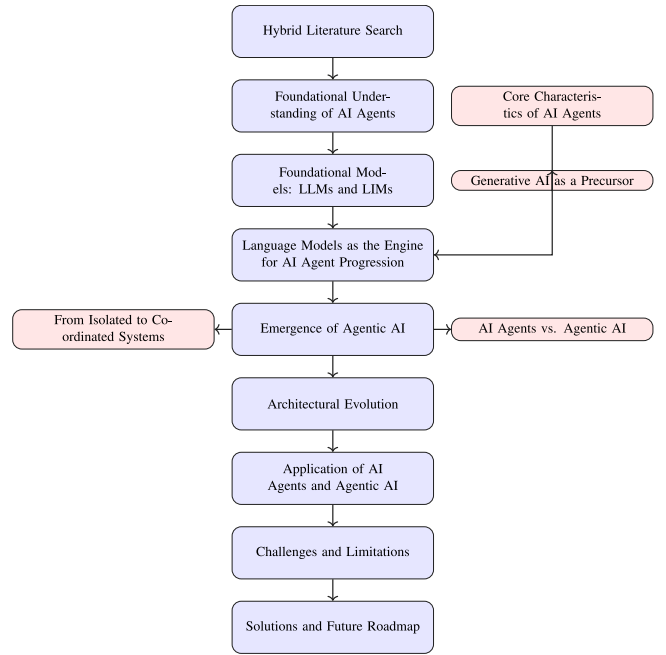


Fig. 3. Methodology pipeline showing the conceptual and architectural progression from foundational AI Agents to emergent Agentic AI. Each stage reflects a distinct layer of conceptual depth, highlighting key mechanisms, comparisons, challenges, and future directions.

Despite growing interest in AI Agents and Agentic AI, existing survey efforts often conflate the two under the broad umbrella of intelligent agents, leading to conceptual ambiguity and misaligned system design. Recent literature either focuses narrowly on tool-augmented LLMs for task automation or broadly discusses multi-agent systems without acknowledging the architectural and functional transition brought forth by Agentic AI. This review offers the first structured taxonomy that formally separates these paradigms across dimensions of autonomy, coordination, interaction, and reasoning scope. By positioning AI Agents as modular, single-entity systems and Agentic AI as orchestrated ecosystems with emergent behaviors, this work addresses a critical gap in distinguishing design principles, deployment strategies, and evaluation criteria. The timing of this distinction is particularly important now, as real-world applications increasingly demand scalable, multi-agent intelligence (e.g., in robotics, healthcare, and scientific discovery), yet developers often lack a clear framework for choosing between isolated agents and collaborative agentic architectures. This review bridges that gap, enabling more precise alignment of system capabilities with task complexity, and informing both research agendas and practical deployments in the post-LLM era.

1.1. Methodology overview

This review adopts a structured, multi-stage methodology designed to capture the evolution, architecture, application, and limitations of AI Agents and Agentic AI. The process is visually summarized in Fig. 3, which delineates the sequential flow of topics and concepts explored in this study. The analytical framework was organized to analyze and present the progression from basic agentic constructs rooted in LLMs to advanced multi-agent orchestration systems. Each step of the review was based on the rigorous synthesis of the literature from across academic sources and AI-powered platforms, enabling a comprehensive understanding of the current landscape and its emerging trends.

The review begins by establishing a *foundational understanding of AI Agents*, examining their core definitions, design principles, and architectural modules as described in the literature. These include components

such as perception, reasoning, and action selection, along with early applications like customer service bots and retrieval assistants. This foundational layer serves as the conceptual entry point into the broader agentic paradigm.

Next, we discuss the role of *LLMs as core reasoning components*, emphasizing how pre-trained language models enable modern AI Agents. This section details how LLMs, through instruction fine-tuning and reinforcement learning from human feedback (RLHF), enable natural language interaction, planning, and limited decision-making capabilities. We also identify their limitations, such as hallucinations, static knowledge, and a lack of causal reasoning.

Building on these foundations, the review proceeds to the *emergence of Agentic AI*, which represents a significant conceptual advancement. Here, we highlight the transformation from tool-augmented single-agent systems to collaborative, distributed ecosystems of interacting agents. This shift is driven by the need for systems capable of decomposing goals, assigning subtasks, coordinating outputs, and adapting dynamically to changing contexts, which are the capabilities that surpass what isolated AI Agents can offer.

The next section examines the *architectural evolution from AI Agents to Agentic AI systems*, contrasting simple, modular agent designs with complex orchestration frameworks. We describe enhancements such as persistent memory, meta-agent coordination, multi-agent planning loops (e.g., ReAct and Chain-of-Thought prompting), and semantic communication protocols. Comparative architectural analysis is supported with examples from platforms like AutoGPT, CrewAI, and LangGraph.

Following the architectural exploration, the review presents an in-depth analysis of *application domains* where AI Agents and Agentic AI are being deployed. The paper discusses four representative application areas for each paradigm. For AI Agents, these include Customer Support Automation, Internal Enterprise Search, Email Filtering and Prioritization, and Personalized Content Recommendation. For Agentic AI, the applications span Multi-Agent Research Assistants, Intelligent Robotics Coordination, Collaborative Medical Decision Support, and Adaptive Workflow Automation. These use cases are examined with respect to system complexity, real-time decision-making, and collaborative task execution.

Subsequently, we address the *challenges and limitations* inherent to both paradigms. For AI Agents, we focus on issues like hallucination, prompt brittleness, limited planning ability, and lack of causal understanding. For Agentic AI, we identify higher-order challenges such as inter-agent misalignment, error propagation, unpredictability of emergent behavior, explainability deficits, and adversarial vulnerabilities. These problems are critically examined with references to recent experimental studies and technical reports.

Finally, the review outlines *potential solutions to overcome these challenges*, drawing on recent advances in causal modeling, retrieval-augmented generation (RAG), multi-agent memory frameworks, and robust evaluation pipelines. These strategies are discussed not only as technical fixes but as foundational requirements for scaling agentic systems into high-stakes domains such as healthcare, finance, and autonomous robotics.

In summary, this methodological structure enables a systematic and comprehensive assessment of the state of AI Agents and Agentic AI. By sequencing the analysis from foundational understanding, to model integration, architectural advancements, applications, and to limitations and potential solutions, the study aims to provide both theoretical clarity and practical guidance to researchers and practitioners navigating this rapidly evolving field.

1.1.1. Search strategy

To develop this review, we implemented a hybrid search methodology combining traditional academic repositories and AI-enhanced literature discovery tools. Specifically, twelve platforms were queried: academic databases such as Google Scholar, IEEE Xplore, ACM Digital

Library, Scopus, Web of Science, ScienceDirect, and arXiv; and AI-powered interfaces including ChatGPT, Perplexity.ai, DeepSeek, Hugging Face Search, and Grok. Search queries incorporated Boolean combinations of terms such as “AI Agents”, “Agentic AI”, “LLM Agents”, “Tool-augmented LLMs”, and “Multi-Agent AI Systems”.

Targeted queries such as “Agentic AI + Coordination + Planning”, and “AI Agents + Tool Usage + Reasoning” were also employed to retrieve papers addressing both conceptual underpinnings and system-level implementations. Literature inclusion was based on their the significance in terms of novelty, empirical evaluation, architectural contribution, and citation impact. The rising global interest in these technologies, as illustrated in Fig. 1 using Google Trends data, underscores the urgency of synthesizing this emerging knowledge space.

2. Foundational understanding of AI Agents

AI Agents can be defined as autonomous software entities engineered for goal-directed task execution within bounded digital environments [22,45]. These agents are defined by their ability to perceive structured or unstructured inputs [46], to reason over contextual information [47,48], and to initiate actions toward achieving specific objectives, often acting as surrogates for human users or subsystems [49]. Unlike conventional automation scripts, which follow deterministic workflows, AI Agents demonstrate reactive intelligence and some level of adaptability, allowing them to interpret dynamic inputs and reconfigure outputs accordingly [50]. Their adoption has been reported across a wide range of application domains, including customer service automation [51,52], personal productivity assistance [53], organizational information retrieval [54,55], and decision support systems [56,57].

A notable example of autonomous AI agents in Anthropic’s “Computer Use” project [computer use](#), which showcases how their Claude model can interact with a computer in much the same way a human would. In this project, Claude is trained to visually interpret what is on a computer screen, control the mouse and keyboard, and navigate through various software applications. This allows Claude to automate repetitive tasks, such as filling out forms or copying data, as well as more complex activities like building and testing software by opening code editors, running commands, and debugging issues. Beyond these structured tasks, Claude can also handle open-ended assignments like conducting online research, gathering and organizing information from multiple sources, and even creating calendar events based on its findings. The key innovation is that Claude operates in an “agent loop”, where it receives a goal, decides on the next action, performs that action, observes the result, and repeats this process until the task is complete. This enables Claude to independently use existing computer tools and interfaces to accomplish a wide range of objectives, making it a powerful example of how autonomous AI agents can automate both routine and complex workflows.

2.0.1. Core characteristics of AI Agents

AI Agents are widely conceptualized as instantiated operational instances of artificial intelligence designed to interface with users, software ecosystems, or digital infrastructures to develop goal-directed behavior [58–60]. These agents are different than general-purpose LLMs in the sense that they exhibit structured initialization, bounded autonomy, and persistent task orientation. While LLMs primarily function as reactive prompt followers [61], AI Agents operate automatically within explicitly defined scopes, engaging dynamically with inputs and producing actionable outputs in real-time environments [62].

Fig. 4 illustrates the three foundational characteristics commonly incorporated by architectural taxonomies and empirical deployments of AI Agents. These characteristics include *autonomy*, *task-specificity*, and *reactivity with adaptation*.

Together, these three characteristics provide a foundational framework for understanding and evaluating AI Agents across deployment scenarios. The remainder of this section elaborates on each characteristic, offering theoretical background and illustrative examples.

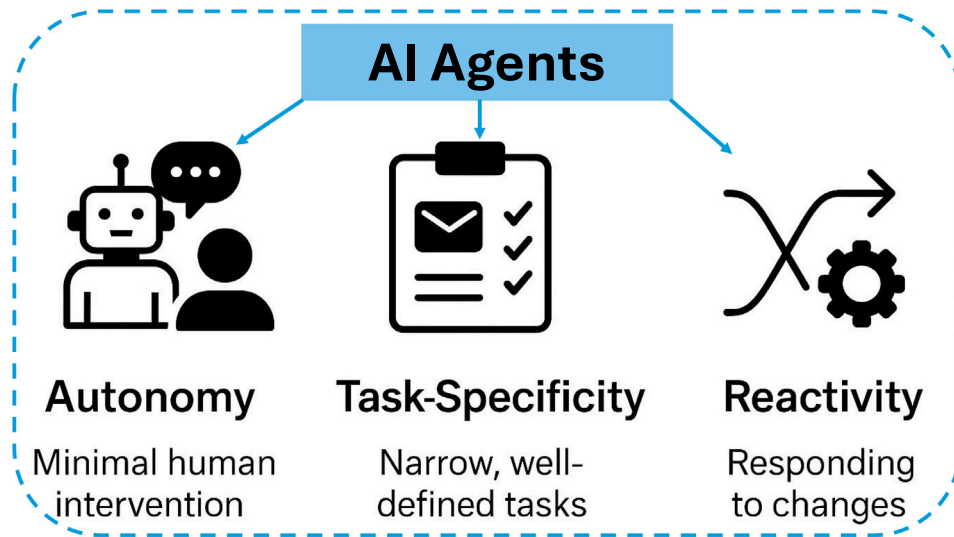


Fig. 4. Illustration of the three core characteristics defining AI Agents: autonomy, task-specificity, and reactivity. These traits form the foundational principles for agent design, enabling goal-oriented, adaptive behaviors within bounded environments and differentiating AI Agents from general-purpose or generative models.

- **Autonomy:** A central feature of AI Agents is their ability to function with minimal or no human intervention after deployment [63]. Once initialized, these agents are capable of perceiving environmental inputs, reasoning over contextual data, and executing predefined or adaptive actions in real-time [25]. Autonomy enables scalable deployment in applications where persistent oversight (human-in-the-loop) is impractical, such as customer support bots or scheduling assistants [52,64].
- **Task-Specificity:** AI Agents are purpose-built for narrow, and well-defined tasks [65,66]. They are optimized to execute repeatable operations within a fixed domain, such as email filtering [67, 68], database querying [69], or calendar coordination [44,70]. This task specialization allows for efficiency, interpretability, and high precision in automating tasks where general-purpose reasoning is unnecessary or inefficient.
- **Reactivity and Adaptation:** AI Agents often include basic mechanisms for interacting with dynamic inputs, allowing them to respond to real-time stimuli such as user requests, external API calls, or state changes in software environments [25,71]. Some systems integrate basic learning capabilities [72] through feedback loops [73,74], heuristics [75], or updated context buffers to refine behavior over time, particularly in settings like personalized recommendations or conversation flow management [76–78].

These core characteristics collectively enable AI Agents to serve as modular, lightweight interfaces between pretrained AI models and domain-specific utility pipelines. Their architectural simplicity and operational efficiency position them as key enablers of scalable automation across enterprise, consumer, and industrial settings. Although there are currently no studies explicitly involving AI Agents integrated with specialized reasoning LLMs, their high usability and performance within constrained task boundaries have made them foundational components in contemporary intelligent system design.

2.0.2. Foundational models: The role of LLMs and LIMs

The progress in AI Agents has been significantly accelerated by the foundational development and deployment of LLMs and LIMs, which serve as the core reasoning and perception engines in contemporary agent systems. These models enable AI agents to interact intelligently with their environments, understand multi-modal inputs, and perform complex reasoning tasks that go beyond hard-coded automation.

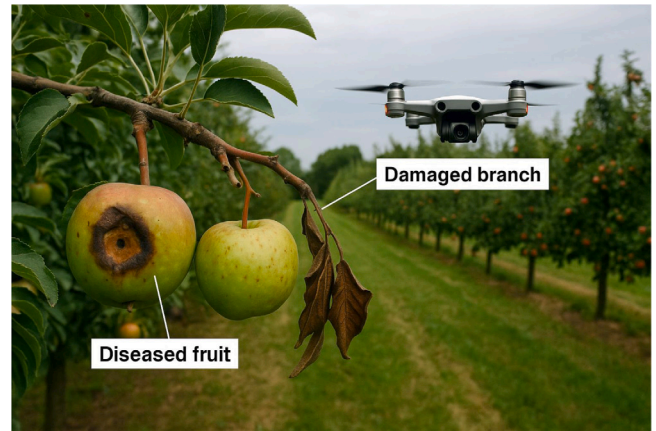


Fig. 5. An AI agent-enabled drone autonomously inspects an orchard, identifying diseased fruits and damaged branches using vision models, and triggers real-time alerts for targeted horticultural interventions.

LLMs such as GPT-4 [79] and PaLM [80] are trained on massive datasets of text from books, web content, and dialogue corpora. These models exhibit emergent capabilities in natural language understanding, question answering, summarization, dialogue coherence, and even symbolic reasoning [81–83]. Within AI Agent architectures, LLMs serve as the primary decision-making engine, allowing the agent to parse user queries, plan multi-step solutions, and generate human-like responses. For instance, an AI customer support agent powered by GPT-4 can interpret customer complaints, query backend systems via tool integration, and respond in a contextually appropriate and emotionally aware manner [84,85].

Large Image Models (LIMs) such as CLIP [86] and BLIP-2 [87] extend the agent's capabilities into the visual domain. Trained on image-text pairs, LIMs enable perception-based tasks including image classification, object detection, and vision-language grounding. These capabilities are increasingly vital for agents operating in domains such as robotics [88], autonomous vehicles [89,90], and visual content moderation [91,92].

For example, as illustrated in Fig. 5 where an autonomous drone agent is tasked with monitoring orchards, a LIM can identify diseased

fruits [93] or damaged branches by interpreting live aerial imagery. Upon detection, the system autonomously triggers predefined intervention protocols, such as notifying horticultural staff or marking the location for targeted treatment without requiring human involvement [25, 63]. This workflow exemplifies the autonomy and reactivity of AI Agents in agricultural environments as highlighted by recent literature indicating the growing sophistication of such drone-based AI Agents. Chitra et al. [94] provides a comprehensive overview of AI algorithms foundational to embodied agents, highlighting the integration of computer vision, SLAM, reinforcement learning, and sensor fusion. These components collectively support real-time perception and adaptive navigation in dynamic environments. Kourav et al. [95] further emphasize the role of natural language processing and LLMs in generating drone action plans from human-issued queries, demonstrating how LLMs support naturalistic interaction and mission planning. Similarly, Natarajan et al. [96] explore deep learning and reinforcement learning for scene understanding, spatial mapping, and multi-agent coordination in aerial robotics. These studies converge on the critical importance of AI-driven autonomy, perception, and decision-making in advancing drone-based agents.

Importantly, LLMs and LIMs are often accessed via inference APIs provided by cloud-based platforms such as OpenAI [Open AI](#), Hugging-Face [Hugging Face](#), and Google Gemini [Google Gemini](#). These services abstract away the complexity of model training and fine-tuning, enabling developers to rapidly build and deploy agents equipped with state-of-the-art reasoning and perceptual abilities. This integrability accelerates prototyping and allows agent frameworks like LangChain [97] and AutoGen [98] to orchestrate LLM and LIM outputs across task workflows. In short, foundational AI models give modern AI Agents their basic understanding of language and scenes. Language models help them reason with words, and image models help them understand pictures; working together, they allow AI Agents to make smart decisions in complex situations.

2.0.3. Generative AI as a precursor

A consistent theme in the literature is the positioning of generative AI as the foundational precursor to agentic intelligence. These systems primarily operate on pre-trained LLMs and LIMs, which are optimized to synthesize multi-modal content including text, images, audio, or code based on input prompts. While highly communicative, generative models fundamentally exhibit reactive behavior: they produce output only when explicitly prompted and do not pursue goals autonomously or engage in self-initiated reasoning [99,100].

Key Characteristics of Generative AI:

- **Reactivity:** As non-autonomous systems, generative models are exclusively input-driven [101,102]. Their operations are triggered by user-specified prompts and they lack internal states, persistent memory, or goal-following mechanisms [103–105].
- **Multi-modal Capability:** Modern generative systems can produce a diverse array of outputs, including coherent narratives, executable code, realistic images, and even speech transcripts. For instance, models like GPT-4 [79], PaLM-E [106], and BLIP-2 [87] demonstrate these capabilities, enabling language-to-image, image-to-text, and cross-modal synthesis tasks.
- **Prompt Dependency and Statelessness:** Although generative systems are stateless in that they do not retain context across interactions unless explicitly prompted [107,108], recent advancements like GPT-4.1 support larger context windows up to 1 million tokens and are better able to utilize that context enabled by the improved long-text comprehension [109]. Their design also lacks intrinsic feedback loops [110], state management [111,112], or multi-step planning a requirement for autonomous decision-making and iterative goal refinement [113, 114].

Despite their remarkable generative fidelity, these systems are constrained by their inability to act upon the environment or manipulate digital tools independently. For instance, they cannot search the internet, parse real-time data, or interact with APIs without human-engineered wrappers or scaffolding layers. As such, they fall short of being classified as true AI Agents, whose architectures integrate perception, decision-making, and external tool-use within closed feedback loops.

The limitations of generative AI in handling dynamic tasks, maintaining state continuity, or executing multi-step plans led to the development of tool-augmented systems, commonly referred to as AI Agents [115]. These systems build upon the language processing backbone of LLMs but introduce additional infrastructure such as memory buffers, tool-calling APIs, reasoning chains, and planning routines to bridge the gap between passive response generation and active task completion. This architectural evolution marks a critical shift in AI system design: from content creation to autonomous task execution [116,117]. The trend from generative systems to AI Agents highlights a progressive layering of functionality that ultimately supports the emergence of agentic behaviors.

2.1. Language models as the engine for AI Agent progression

The emergence of AI Agent as a transformative paradigm in artificial intelligence is closely tied to the evolution and repurposing of large-scale language models such as GPT-3 [118], Llama [119], T5 [120], Baichuan 2 [121] and GPT3mix [122]. A substantial and growing body of research shows that the advancement, from reactive generative models to autonomous, goal-directed agents is driven by the integration of LLMs as core reasoning engines within dynamic agentic systems. These models, originally trained for natural language processing tasks, are increasingly embedded in frameworks that require adaptive planning [123,124], real-time decision-making [125, 126], and environment-aware behavior [127].

2.1.1. LLMs as core reasoning components

LLMs such as GPT-4 [79], PaLM [80], Claude [Claude 3.5 Sonnet](#), and LLaMA [119] are pre-trained on massive text corpora using self-supervised objectives and fine-tuned using techniques such as Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) [128,129]. These models encode rich statistical and semantic knowledge, allowing them to perform tasks like inference, summarization, code generation, and dialogue management. However, in agentic contexts, their capabilities extend beyond response generation. They function as cognitive engines that interpret user goals, formulate and evaluate possible action plans, select the most appropriate strategies, leverage external tools, and manage complex, multi-step workflows.

Recent work identifies these models as central to the architecture of contemporary agentic systems. For instance, AutoGPT [38] and BabyAGI [BabyAGI](#) use GPT-4 as both a planner and executor: the model analyzes high-level objectives, decomposes them into actionable subtasks, invokes external APIs as needed, and monitors progress to determine subsequent actions. In such systems, the LLM operates in a loop of prompt processing, state updating, and feedback-based correction, closely emulating autonomous decision-making.

2.1.2. Tool-augmented AI Agents: Enhancing functionality

To overcome limitations inherent to generative-only systems such as hallucination, static knowledge cutoffs, and restricted interaction scopes, researchers have proposed the concept of tool-augmented AI Agents [130] such as Easytool [131], Gentopia [132], and ToolFive [133]. These systems integrate external tools, APIs, and computation platforms into the agent's reasoning pipeline, allowing for real-time information access, code execution, and interaction with dynamic data environments.

Tool Invocation. When an agent identifies a need that cannot be addressed through its internal knowledge such as querying a current stock price, retrieving up-to-date weather information, or executing a script, it generates a structured function call or API request [134,135]. These calls are typically formatted in JSON, SQL, or Python dictionary, depending on the target service, and routed through an orchestration layer that executes the task.

Result Integration. Once a response is received from the tool, the output is parsed and reincorporated into the LLM's context window. This enables the agent to synthesize new reasoning paths, update its task status, and decide on the next step. The ReAct framework [136] exemplifies this architecture by combining reasoning (Chain-of-Thought prompting) and action (tool use), with LLMs alternating between internal cognition and external environment interaction. A prominent example of a tool-augmented AI agent is ChatGPT, which, when unable to answer a query directly, autonomously invokes the Web Search API to retrieve more recent and relevant information, performs reasoning over the retrieved content, and formulates a response based on its understanding [137].

2.1.3. Illustrative examples and emerging capabilities

Tool-augmented LLM-powered AI Agents have demonstrated potentials across a range of applications. In AutoGPT [38], the agent may plan a product market analysis by sequentially querying the web, compiling competitor data, summarizing insights, and generating a report. In a coding context, tools like GPT-Engineer combine LLM-driven design with local code execution environments to iteratively develop software artifacts as output produced during the development process, including source code, .exe files, documentation and configurations [138,139]. In research domains, systems like Paper-QA [140] utilize LLMs to query vectorized academic databases, grounding answers in retrieved scientific literature to ensure factual integrity.

These capabilities have opened pathways for more robust behavior of AI Agents such as long-horizon planning, cross-tool coordination, and adaptive learning loops. Nevertheless, the inclusion of tools also introduces new challenges in coordination complexity, error propagation, and context window limitations, which are all active areas of research. The progression toward AI Agents is inseparable from the strategic integration of LLMs as reasoning engines and their augmentation through structured utilization of external tools like search engines and APIs. This synergy transforms static language models into dynamic cognitive agents capable of perceiving, planning, acting, and adapting, thus setting the stage for multi-agent collaboration, persistent memory, and scalable autonomy, the characteristics of the Agentic AI systems.

As an example, Fig. 6 illustrates a representative use-case: a news query agent that performs real-time web search, summarizes retrieved documents, and generates an articulate, context-aware answer. Such workflows have been demonstrated in implementations using LangChain, AutoGPT, and OpenAI function-calling architectures.

3. The emergence of Agentic AI from AI Agent foundations

While AI Agents represent a significant leap in artificial intelligence capabilities, particularly in automating narrow tasks through tool-augmented reasoning, recent literature identifies notable limitations that constrain their scalability in complex, dynamic, multi-step, and/or cooperative scenarios [141–143]. These constraints have catalyzed the development of a more advanced paradigm: *Agentic AI*. This emerging class of systems extends the capabilities of traditional AI Agents by enabling multiple intelligent entities to collaboratively pursue goals through structured communication [144–146], shared memory [147,148], and dynamic role assignment [22].

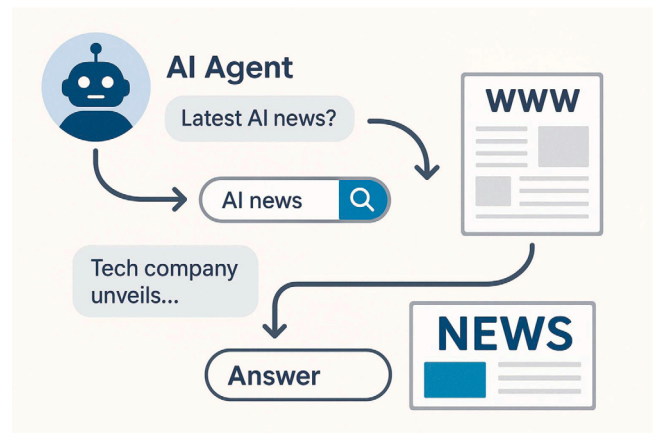


Fig. 6. Illustration of an AI Agent workflow for real-time news search: the agent receives a user query, performs web retrieval via external tools, summarizes retrieved articles, and generates a coherent, context-aware response. This process demonstrates autonomous tool use, reasoning, and sequential task execution.

3.0.1. Conceptual leap: From isolated agents to coordinated systems

AI Agents, as explored in prior sections, integrate LLMs with external tools and APIs to execute narrowly scoped operations such as responding to customer queries, performing document retrieval, or managing schedules. However, as use cases increasingly demand context retention, task interdependence, and adaptability across dynamic environments, the single-agent model proves insufficient [149,150].

Agentic AI systems represent an emergent class of intelligent architectures in which multiple specialized agents collaborate to achieve complex, high-level objectives utilizing collaborative reasoning and multi-step planning [41]. As defined in recent frameworks, these systems are composed of modular agents each tasked with a distinct subcomponent of a broader goal and coordinated through either a centralized orchestrator or a decentralized protocol [24,145]. This structure signifies a conceptual departure from the individual, reactive behaviors typically observed in single-agent architectures, toward a form of system-level intelligence characterized by dynamic inter-agent collaboration.

A key enabler of this paradigm is **goal decomposition**, wherein a user-specified objective is automatically parsed and divided into smaller, manageable tasks by planning agents [44]. These subtasks are then distributed across the agent network. **Multi-step reasoning and planning** mechanisms facilitate the dynamic sequencing of these subtasks, allowing the system to adapt in real time to environmental changes or partial task failures. This agentic architecture ensures robust task execution even under uncertainty [22].

Inter-agent communication is mediated through **distributed communication channels**, such as asynchronous messaging queues, shared memory buffers, or intermediate output exchanges, enabling coordination without necessitating continuous central oversight [22,151]. Furthermore, **reflective reasoning and memory systems** allow agents to store context across multiple interactions, evaluate past decisions, and iteratively refine their strategies [152]. Collectively, these capabilities enable Agentic AI systems to exhibit flexible, adaptive, cooperative, and collaborative intelligence that exceeds the operational limits of individual agents.

A widely accepted conceptual illustration in the literature delineates the distinction between AI Agents and Agentic AI through the analogy of smart home systems. As depicted in Fig. 7, the left side represents a traditional AI Agent in the form of a smart thermostat. This standalone agent receives a user-defined temperature setting and autonomously controls the heating or cooling system to maintain the target temperature. While it demonstrates limited autonomy such as learning

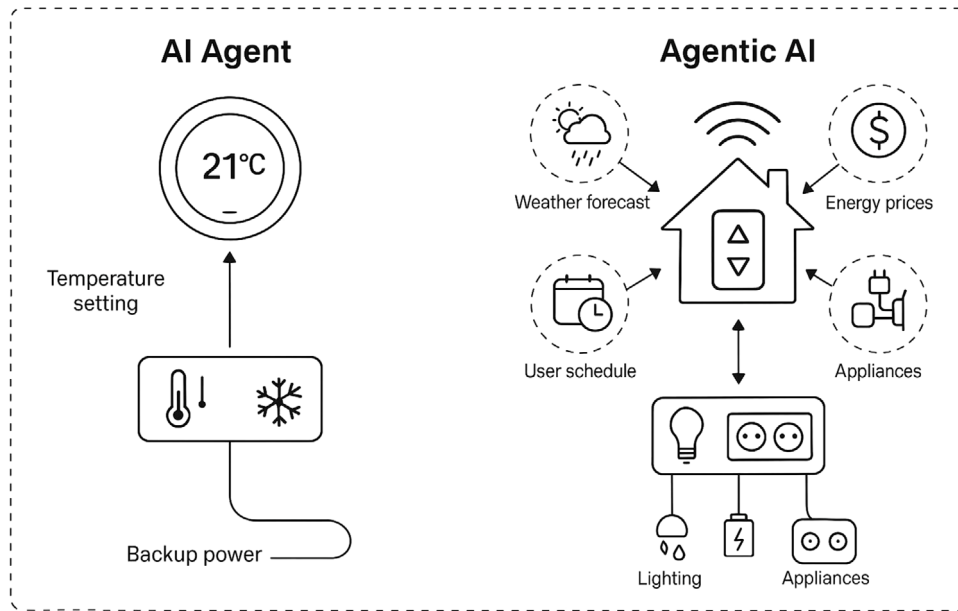


Fig. 7. Comparative illustration of AI Agent versus Agentic AI. The left side shows a basic AI Agent managing a single task — such as maintaining a thermostat’s temperature setting based on user input. In contrast, the right side represents an Agentic AI system coordinating multiple agents to manage household automation tasks including temperature regulation, weather forecasting, energy pricing, appliance scheduling, and security monitoring. This highlights the architectural leap from narrow, reactive control to dynamic, multi-agent orchestration supporting adaptive, goal-driven intelligent environments.

user schedules or reducing energy usage during absence, it operates in isolation, executing a singular, well-defined task without engaging in broader environmental coordination or goal inference [25,63].

In contrast, the right side of Fig. 7 illustrates an Agentic AI system embedded in a comprehensive smart home ecosystem. Here, multiple specialized agents interact synergistically to manage diverse aspects such as weather forecasting, daily scheduling, energy pricing optimization, security monitoring, and backup power activation. These agents are not just reactive modules; they communicate dynamically, share memory states, and collaboratively align actions toward a high-level system goal (e.g., optimizing comfort, safety, and energy efficiency in real-time). For instance, a weather forecast agent might signal upcoming heatwaves, prompting early pre-cooling via solar energy before peak pricing hours, as coordinated by an energy management agent. Simultaneously, the system might delay high-energy tasks or activate surveillance systems during occupant absence, integrating decisions across domains. This figure embodies the architectural and functional leap from task-specific automation to adaptive, orchestrated intelligence. The AI Agent acts as a deterministic component with limited scope, while Agentic AI reflects distributed intelligence, characterized by goal decomposition, inter-agent communication, and contextual adaptation, demonstrating key characteristics of the modern agentic AI frameworks.

3.0.2. Key differences between AI Agents and Agentic AI

To systematically capture the evolution from Generative AI to AI Agents and further to Agentic AI, we structure our comparative analysis around a foundational taxonomy where Generative AI serves as the baseline. While AI Agents and Agentic AI systems represent increasingly autonomous and interactive systems, both paradigms utilize generative architectures as their foundations, especially LLMs and LIMs. Consequently, each comparative table in this subsection includes Generative AI as a reference column to highlight how agentic behavior builds on and then diverges from generative AI foundations.

A set of basic distinctions between AI Agents and Agentic AI systems, particularly in terms of scope, autonomy, architectural composition, coordination strategy, and operational complexity, are synthesized in Table 1 and Table 10 which was derived from close analysis of

prominent frameworks such as AutoGen [98] and ChatDev [153]. This comparison provides a multi-dimensional view of how single-agent systems transition into coordinated, multi-agent ecosystems. Through the perspective of generative capabilities, we trace the increasing sophistication in planning, communication, and adaptation that characterizes the shift toward Agentic AI systems.

While Table 1 delineates the foundational and operational differences between AI Agents and Agentic AI systems, a more granular taxonomy is required to understand how these paradigms emerge from and relate to broader generative AI frameworks. Specifically, the conceptual and cognitive progression from static Generative AI systems to tool-augmented AI Agents, and further to collaborative Agentic AI ecosystems, necessitates an integrated comparative framework. This transition is not merely structural but also functional encompassing how initiation mechanisms, memory use, learning capacities, and orchestration strategies evolve across the agentic spectrum. Moreover, recent studies suggest the emergence of hybrid paradigms such as “Generative Agents”, which blend generative modeling with modular task specialization, further complicating the agentic AI landscape. In order to capture these nuanced relationships, Table 2 synthesizes the key conceptual and cognitive dimensions across four archetypes: Generative AI, AI Agents, Agentic AI systems, and inferred Generative Agents. By positioning Generative AI as a baseline technology, this taxonomy highlights the scientific, structural and application continuum that spans from passive content generation to interactive task execution and finally to autonomous, multi-agent orchestration. This multi-tiered perspective is critical for understanding both the current capabilities and future trends of agentic intelligence across theoretical and applied domains.

To further operationalize the distinctions outlined in Table 1, Tables 2 and 3 extend the comparison between agent paradigms to encompass a broader spectrum of paradigms including AI agents and agentic AI. Table 3 presents key architectural and behavioral attributes that highlight how each paradigm differs in terms of primary capabilities, planning scope, interaction style, learning dynamics, and evaluation criteria. As can be seen from the tables, AI Agents are optimized for discrete task execution with limited planning horizons and rely on supervised or rule-based learning mechanisms. In contrast,